Bio-Hack 2026

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Problem Statement

Demonstrate a need for innovative enteral feeding devices that are safer, easier to use, more personalized, and accessible across different care environments.

Inspiration

Studies have shown that elderly individuals are at higher risk for complications, commonly including clogging.

- ~25-45% of long-term enteral feeding tubes suffer from occlusions (Galindo-Ciocon, D[4])
- $200-$800 dollar cost for tube replacement if clogs cause internal damages (CCG Editorial [2])

We wanted to devise an innovation that would help combat that risk, especially for the elderly with restricted mobility or healthcare while also supporting a low maintenance lifestyle.

Standard Shift

- ”Active vs Reactive” Methodology shift
  -Current standards (syringe flushing, water, enzymes, or tube swaps) are 100% reactive after full blockage occurs

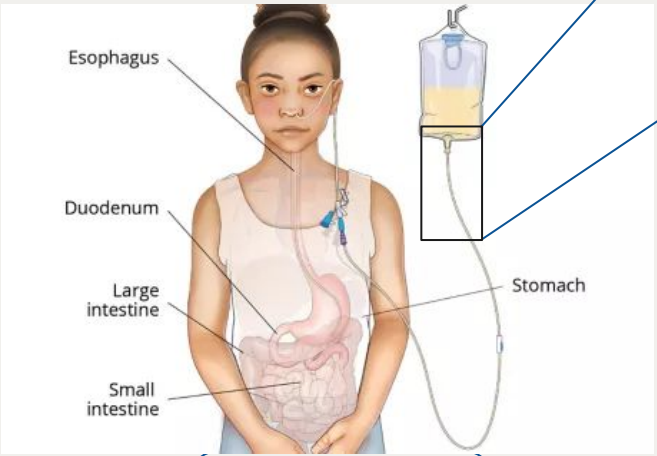
  -Our standards focus on a prioritization of an external system to not modify fluid flow or sterility.

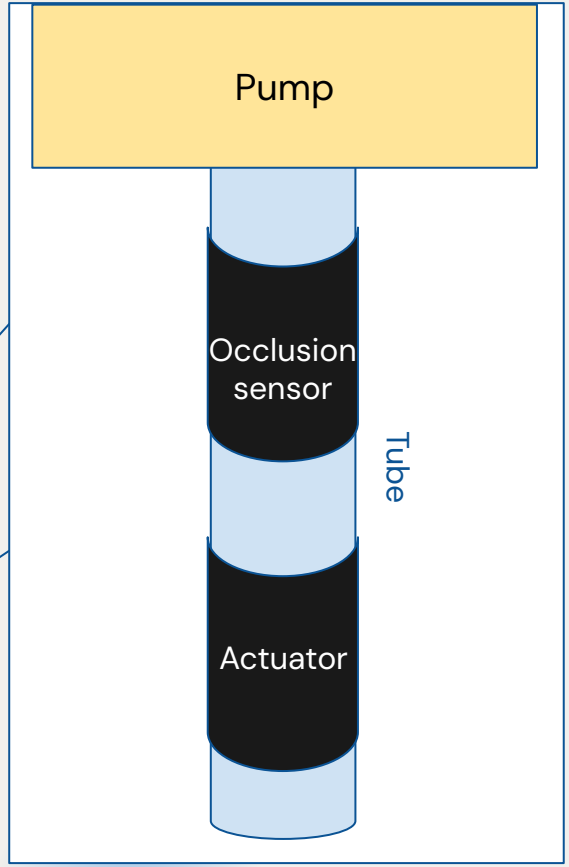
What It Does

An external motor vibrates the feeding tube to create a high-frequency mechanical agitation directly to the trapped material. This agitation allows for automatic dispersion of the material or help assist liquids penetrate the material.

How It Works

The vibration is automatically triggered and adjusted by an occlusion strain sensor, which is a non-invasive monitoring device that uses strain-sensing technology to detect blockages.





Implementation

This product can be initially implemented into hospital care with professional oversight before being involved in elderly care facilities. This allows for the product to be monitored closely before giving more flexible freedom to elderly care facilities that have less oversight.

Advantages

Immobility

The elderly who struggle with mobility can benefit from the automatic feature.

Memory Loss

Forgetting proper procedures can be less dangerous.

Comfortability

Removing clogs through comfortable means.

Independency

Potential for the need of Less caretakers.

Cost Effective

Potential lower hospital and medical expenses by reducing visits.

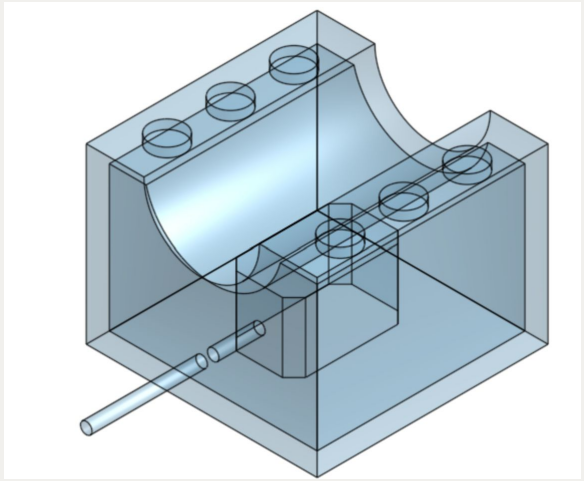
Safety

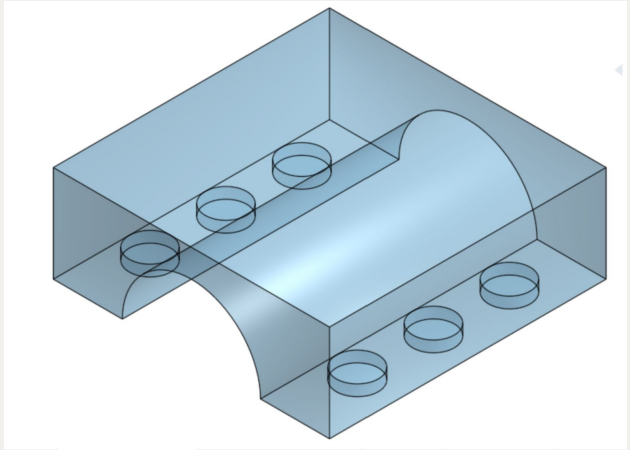
If the user doesn't react to emergency safety mechanisms, the automatic vibrations can potentially clear a clog.

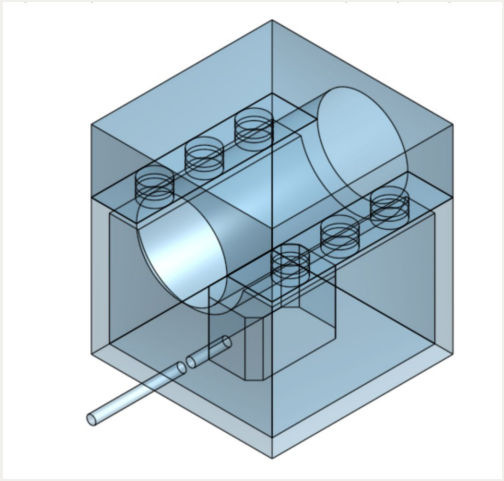
Our Representation

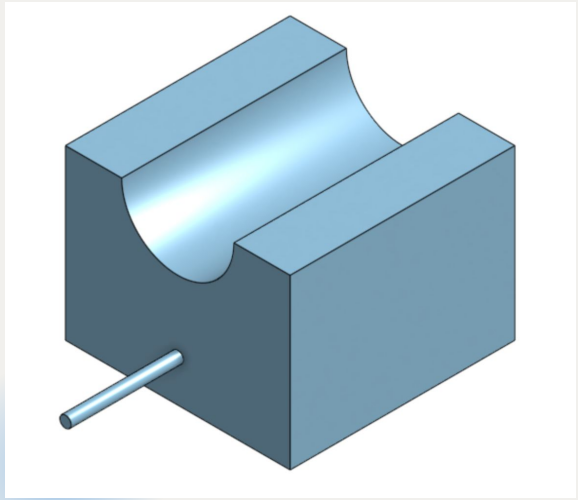
CAD

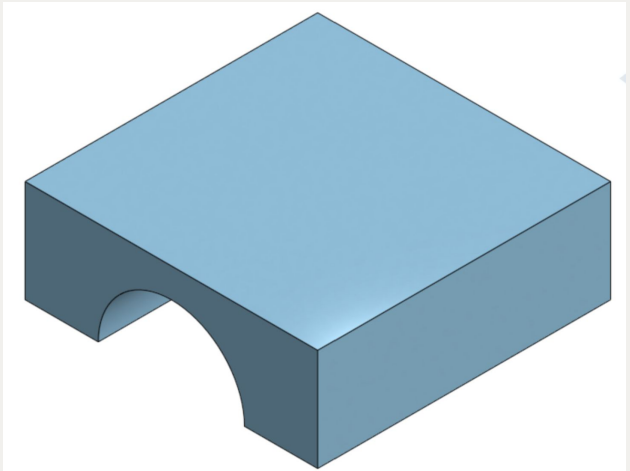
Occlusion Sensor

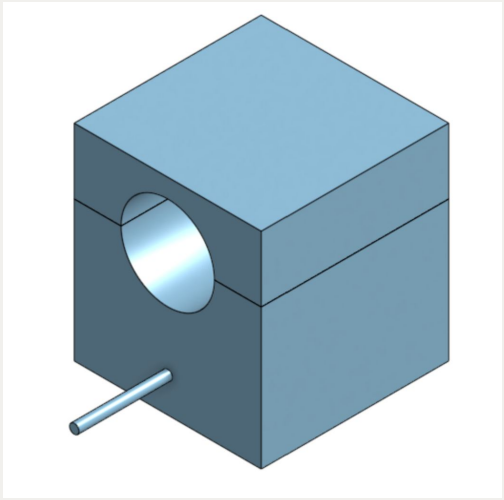




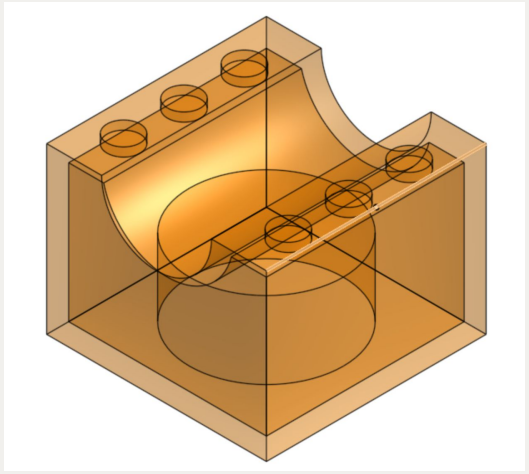


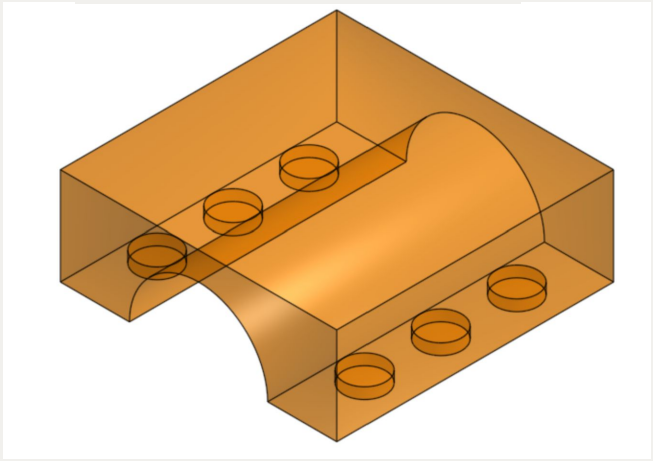


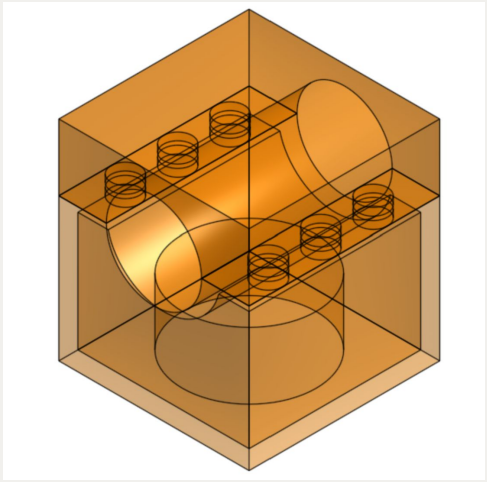


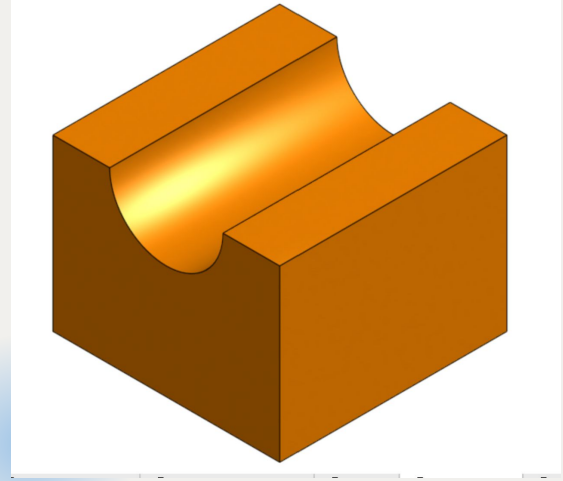


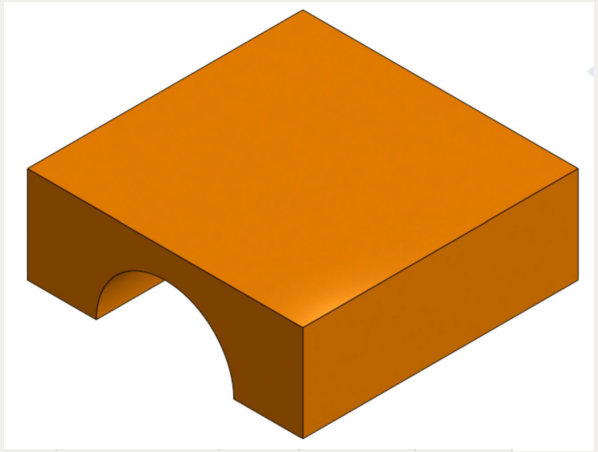
Actuator

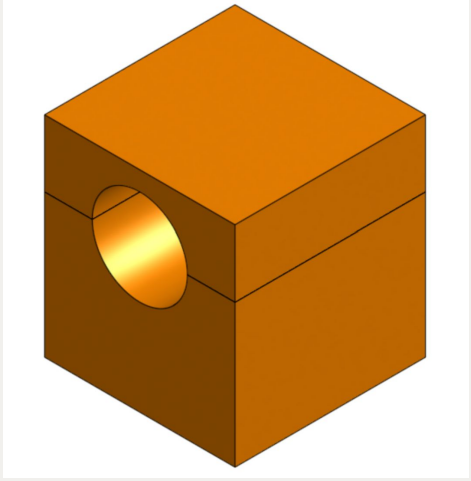












A Closed-Loop Approach to Occlusion Detection

- Detect
  - -Constant monitoring for signs of an obstruction
- Respond
  - -Activate external vibration when detection confidence is sufficiently high
- Confirm
  - -Monitor the obstruction until clearance

Our simulation models our proposed system as a closed-loop process, rather than a one-time intervention.

Real-Time Feedback & Signal Analytics

- Threshold Limits
  -Tracks voltage variations against (non-clinical) warning (30%) and critical (75%) thresholds
- < 5.5 s Occlusion Clearance Response
  Vibration engages automatically when confidence exceeds acceptable threshold
- Hysteresis Auto-Shut Off
  -Actuator turns off once obstruction level drops below a 0.08 clearance threshold to prevent unnecessary mechanical wear or patient disturbance

Code Showcase

```
sensor_problem = (
    self.filtered_sensor_signal
    > self.sensor_warning_threshold
)

## Require several abnormal samples

if sensor_problem:

    self.detection_counter += 1

else:

    self.detection_counter = max(
        0,
        self.detection_counter - 1
    )

if (
    self.filtered_sensor_signal <= self.sensor_warning_threshold
):
    self.detection_confidence = 0.0
else:
    signal_range = (
        self.sensor_critical_threshold
        - self.sensor_warning_threshold
    )
    signal_confidence = (
        self.filtered_sensor_signal
        - self.sensor_warning_threshold
    ) / signal_range
    signal_confidence = max(
        0.0,
        min(1.0, signal_confidence)
    )

    persistence_confidence = min(
        1.0,
        self.detection_counter
        / self.required_detection_samples
    )
## both a strong signal and persistence are required for high confidence

    self.detection_confidence = (
        signal_confidence
        *persistence_confidence
    )
```

```
## CLEARANCE CONFIRMATION


if (
    self.occlusion_detected
    and self.occlusion < 0.08
    and self.filtered_sensor_signal < 20
):

    self.occlusion_detected = False

    self.vibration_active = False

    self.target_occlusion = 0.0

    self.detection_counter = 0

    self.log(
        "CLEARANCE CONFIRMED: "
        "occlusion sensor returned toward baseline"
    )
```

marygreen0325/OSU-BioHack2026-Team9: 10/03/2026

```
## Arbitrary simulation parameter.
##
## This does NOT represent a real clinical
## actuator setting.

clearance = (
    self.clearance_rate
    * intensity
    * dt
)

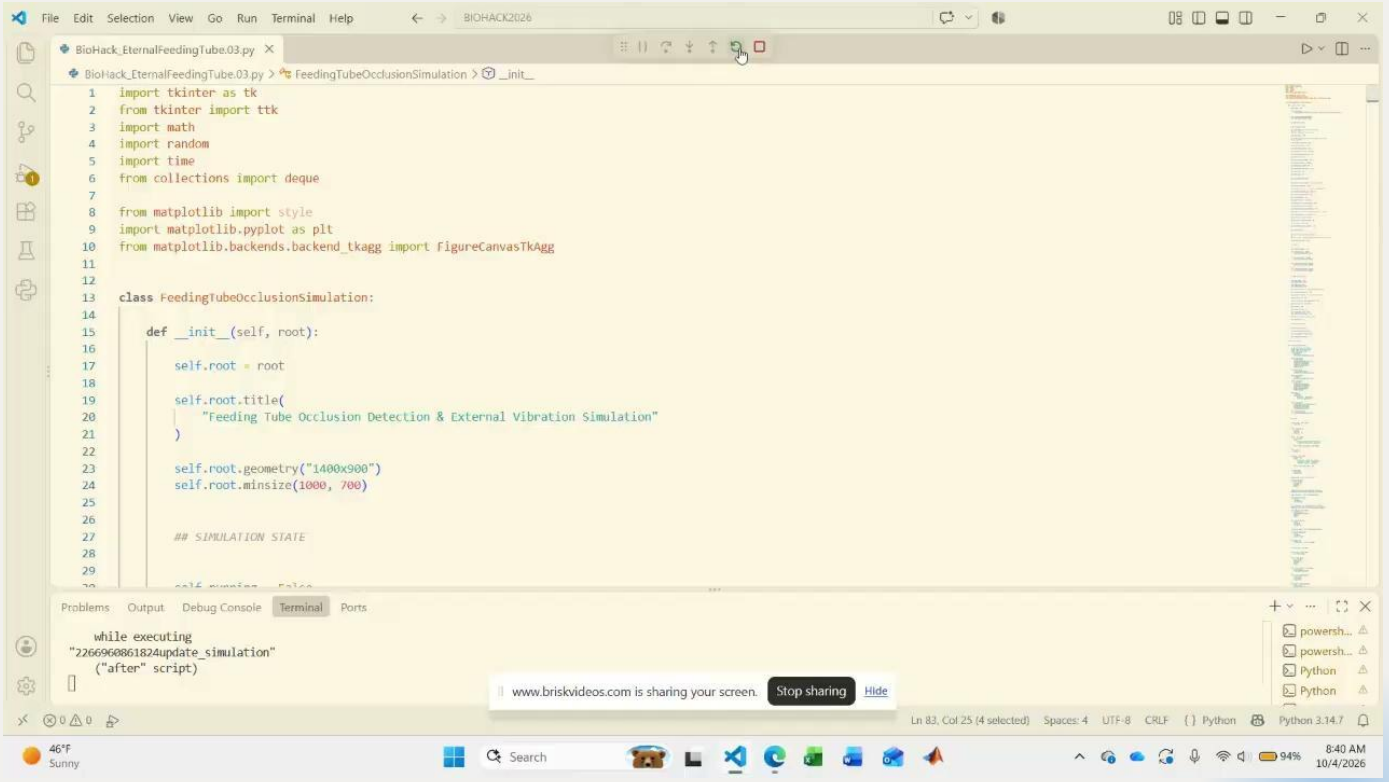
variation = random.uniform(
    0.7,
    1.15
)

clearance *= variation

self.occlusion -= clearance

self.occlusion = max(
    0,
    self.occlusion
)
```

Simulation Demonstration



Closed-Loop System Logic

Occlusion sensor	ABNORMAL
Signal processing	ABNORMAL
Occlusion detection	CLEAR
Controller	IDLE
External actuator	OFF

DETECTION CONFIDENCE

47%

CONTROL SEQUENCE

1. Occlusion sensor continuously samples
2. Sensor signal is filtered
3. Abnormal signal is detected
4. Detection confidence increases
5. Controller activates actuator
6. Simulated obstruction decreases
7. Sensor confirms clearance

Challenges

- Choosing between internal and external designs.
  Sensors and Actuators
- Type of Sensor.... sound, thermal, pressure, etc.
- Hygiene for the user.
- Reliable sensor data analytics

What we've learned:

Creating solutions to problems isn't always about being revolutionary, but also about making practical solutions for helping people live easier.

Thank you.

Bibliography

[1] Alivizatos, V. (2012). *Feeding Tube-related Complications and Problems in Patients Receiving Long-term Home Enteral Nutrition*. file:///C:/Users/maryg/Downloads/Feeding_tube-related_complications_and_problems_in%20(1).pdf

[2] ColonoscopyCostGuide Editorial Team. (2026, May 4). *"PEG Feeding Tube Placement Cost: What a Gastrostomy Tube Runs in 2026."* Colonoscopy Cost Guide; "Colonoscopy Cost Guide." https://colonoscopycostguide.com/guides/peg-feeding-tube-placement-cost/

[3] FUTEK Advanced Sensor Technology. (2026). Futek.Com. https://www.futek.com/applications/occlusion-sensor

[4] Galindo-Ciocon, Daisy J. 1993. "TUBE FEEDING: Complications Among the Elderly." *Journal of Gerontological Nursing* 19 (6): 17–22. https://doi.org/10.3928/0098-9134-19930601-07.

[5] Zhang, Yingying, Zhina Gong, Jianzheng Cai, Weixia Yu, Yinuo Dai, and Haifang Wang. 2025. "Incidence of Dysphagia-Related Safety Incidents in Older Adults Across Feeding Methods: A Systematic Review and Meta-Analysis." *The Journal of Nutrition, Health and Aging* 29 (5): 100522. https://doi.org/10.1016/j.jnha.2025.100522.